



Aziz Shameem
Electrical Engineering
Machine Intelligence and Data Science
Indian Institute of Technology Bombay

20D070020
Dual Degree (B.Tech. + M.Tech.)
Gender: Male
DOB: 11/02/2002

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	9.68
Intermediate	CBSE	City International School	2020	95.20%
Matriculation	ICSE	Vibgyor High International School	2018	97.00%

SCHOLASTIC ACHIEVEMENTS

- Recipient of the **CMiNDS Fellowship** (awarded to 2 students transitioning to the CMiNDS program) [2023]
- Recipient of the **Summer@EPFL Travel Scholarship** (awarded to 1.6% of the applicants) [2024]
- Ranked **1st** in Electrical Engineering Dept. (Dual Degree) out of a batch of 100 students for two years [2022-2024]
- Awarded **Institute Academic Prize** for exceptional performance in academics for **two consecutive years** [2022,2023]
- Conferred with **AP grades** in **Advanced ML, Linear Algebra, Electronic Devices** (awarded to <2% of students)
- Secured **AIR 62** (amongst 10.5 lakh) in **JEE Mains**, and **AIR 665** (amongst 2.5 lakh) in **JEE Advanced** [2020]
- Publications : **Speeding up NAS with Adaptive Subset Selection** at **AutoML, 2024** [2024]

INTERNSHIPS

Modem Systems Engineering Intern

[May'23-Jul'23]

QUALCOMM TECHNOLOGIES INC.

- Undertook extensive research on enhancing **information-deficit channel estimation** methods for **5G**
- Implemented existing as well as proposed technology in a **python simulator** and verified theoretical results
- Implemented specific **channel models** in the python simulator, to be used for future research and development

RESEARCH EXPERIENCES

Variational Auto-Encoders and Diffusions Models

[May'24-Present]

PROF ALEXANDRE ALAHI | VISUAL INTELLIGENCE FOR TRANSPORTATION(VITA) LABORATORY, EPFL

- Performed extensive analysis of the **similarities and differences** between two types of Generative Frameworks - **VAEs** and **Diffusion Models**
- Designed experiments to elucidate the **limitations** of VAEs, and suggested **improvements/methods** to overcome them
- Demonstrated better generations by VAEs using the improvements recognized **synthetic** as well as **benchmark datasets**
- Working on incorporating the methods for improving **conditional image generation** using VAEs
- Performing analysis of the VAE and Diffusion model used in **Stable Diffusion v2**, to further recognize areas of improvements

Permutation-Aided Regularization based Sequence Matching (PARSEQ)

[Jun'23-Present]

PROF ABIR DE | COMPUTER SCIENCE AND ENGINEERING, IITB

- Developed a **theoretical framework** based on **lagrangians** for **fast, approximate, constrained sequence matching**
- Tested the developed framework **computationally**, and obtained results on datasets of several **modalities**
- Compared the developed method with several baselines, and demonstrated upto **30x speedup** against the best prior work

Neural Architecture Search

[May'23-Sept'23]

PROF GANESH RAMAKRISHNAN | DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

- Enhanced **Pruning-based Neural Architecture search** by incorporating proven pruning techniques
- Incorporated **adaptive subset selection** to enhance the efficient use of data and obtained competitive results in the same

KEY PROJECTS

Toxicity Detection in Large Language Models

[Apr'24-May'24]

COURSE PROJECT | PROF PUSHPAK BHATTACHARYA, DEPT. OF COMPUTER SCIENCE AND ENGINEERING

- Trained a **text toxicity classifier** using **Recurrent Neural Nets, LSTMs** and **Transformers**
- Engineered additional features** from the analysis of textual data in an effort to improve classification of toxicity in texts
- Utilised the trained classifier network as a **filter** to mitigate toxicity in the output of **Large Language Models**

Auto-Grad Engine and GPT development

[May'23-Jun'23]

SELF PROJECT

- Built an **Auto-Grad Engine** having **pytorch-like API** from scratch and demonstrated its working on a test bed
- Performed extensive study on **innate problems** in neural networks pertaining to **gradients and dead neurons**
- Implemented **initialization and normalization based methods** to bypass problems and demonstrated better results
- Developed a working **GPT-like architecture** from scratch and generated recognizable **Shakespearean text**

Semi Supervised Data Programming : SPEAR

[Apr'23-May'23]

COURSE PROJECT | PROF GANESH RAMAKRISHNAN, CENTRE FOR MACHINE LEARNING AND DATA SCIENCE

- Contributed to the SPEAR package by incorporating a previously studied **Bilevel Optimisation** framework
- Incorporated **modular classes** capable of identifying **Labelling Functions** from an input datasets
- Constructed modular functions for **reweighing of Auto-Induced Labeling Functions**, and displayed their performance

Correlated Multi-Armed Bandits

[Mar'23-Apr'23]

COURSE PROJECT | PROF JAYAKRISHNAN NAIR, PROF D. MANJUNATH, ELECTRICAL ENGINEERING DEPARTMENT

- Performed an extensive literature review on the formulation of the **correlated multi-armed Bandit** problem
- Applied an existing framework for the **exploitation of correlation** between bandits, on different algorithms and demonstrated **substantial improvements** in the same

Machine Learning for COVID-19 Data Analysis

[Oct'21-Nov'21]

COURSE PROJECT | PROF AMIT SETHI, CENTRE FOR MACHINE LEARNING AND DATA SCIENCE

- Performed extensive **Exploratory Data Analysis** on **COVID-19** in India and several other countries across two years
- Used **Linear, Polynomial, Lasso, Ridge Regression** models to predict future COVID-19 related casualties
- Implemented **Logistic Regression, SVM, NN, RF** and **GBC** to predict need of ICU admission based on medical factors

Expense Tracker : An Application to track, store and plot timely User Expenses

[May'21-Jul'21]

SELF PROJECT

- Created an interactive **Front-End** for the project to enable **account creation** and **data entry** into the system using tkinter
- Used Python Libraries **Pandas** and **Matplotlib** to create a **Back-End** to store and revert data as and when required

POSITIONS OF RESPONSIBILITY

Machine Learning Engineer

[Jun'22-Apr'23]

DATA ANALYSIS AND VISUALISATION (DAV) TEAM, IITB

- Led a team to **successful completion and publication** (On the official public handle) of analysis on **Indian Establishments**
- Performed analysis on **Grading and Semester Exchange** statistics of the institute and oversaw its **publication on LinkedIn**
- Implemented the **YOLO version-1** framework from scratch as part of a study project and gave a presentation on the same
- Represented team at a kaggle competition which involved **data pre-processing, model design and training** on NLP tasks
- Delivered a comprehensive presentation on **Loss Functions for Image Restoration** and got recognized for the same

Mentor : Introduction to Machine Intelligence

[Apr'22 - Jul'22]

SUMMER OF CODE, 2022

- Prepared and distributed **comprehensive resources** to help beginners understand and appreciate concepts in ML
- Assisted beginners in building trivial **AI systems** using **Reinforcement Learning** in an effort to generate interest in the field

TECHNICAL SKILLS

Programming : Python(Pytorch, tensorflow, keras), C/C++, VHDL, MATLAB, HTML, CSS, JS, Julia, Assembly

Softwares : Git, Vim, L^AT_EX, MATLAB, Quartus Prime, Kicad, Arduino, MySQL, Pycharm, Anaconda, Unity

MISCELLANEOUS

TASHIP	• Complex Analysis • Linear Algebra	• Introduction to Machine Learning • Programming for Data Science
CHESS	• Secured First Place in 9-Up, a tournament organized by Hostel-9 sports council • Represented Hostel 9 in the General Championships at IITB for two consecutive years • Received a trophy for Honourable Mention at the 1st Tarabai Shinde Chess Tournament	
CODING	• Stood third in an Algorithmic Trading hackathon organised by Web and Coding club, IITB	
TENNIS	Represented Hostel 9 in the General Championships at IITB for two consecutive years	